ADS1298 #1

Electrodes

E1 E2 E3 E4 E5 E6 E7 E8

R1 2KΩ R2 2KΩ R3 2KΩ R4 2KΩ R5 2KΩ R6 2KΩ R7 2KΩ R8 2KΩ R9 2KΩ R10 2KΩ R11 2KΩ R12 2KΩ R13 2KΩ R14 2KΩ R15 2KΩ R16 2KΩ

C1 C2 C3 C4 C5 C6 C7 C8 C9 C10 C11 C12 C13 C14 C15 C16 C17 C18 C19 C20 C21 C22 620pF

C6 0.1uF C7 10uF C8 0.1uF C9 10uF C19 0.1uF C20 10uF

C23 22uF C24 1uF C25 1uF C26 1uF

+3.3V AGND

U1 ADS1298xPAG

IN1P IN1N IN2P IN2N IN3P IN3N IN4P IN4N IN5P IN5N IN6P IN6N IN7P IN7N IN8P IN8N

AVDD AVDD1 DVDD AVSS AVSS1 DGND

WCT RLDOUT RLDIN RLDINV RLDREF

TESTP_PACE_OUT1 TESTP_PACE_OUT2

GPIO1 GPIO2 GPIO3 GPIO4

DAISY_IN SCLK DIN CS DOUT

CLK CLKSEL START DRDY PWDN RESET RESV1

VCAP1 VCAP2 VCAP3 VCAP4 VREFP VREFN

VCAP1_ADC1 VCAP2_ADC1 VCAP3_ADC1 VCAP4_ADC1 VREFP_ADC1

SCLK MOSI CS_ADC1 MISO DRDY_ADC1 PWDN RESET

Unused, shorted

U1 CLK_EN=1 required

ADC_CLK +3.3V

ADS1298 #2

Electrodes

E9 E10 E11 E12 E13 E14 E15 E16

R17 2KΩ R18 2KΩ R19 2KΩ R20 2KΩ R21 2KΩ R22 2KΩ R23 2KΩ R24 2KΩ R25 2KΩ R26 2KΩ R27 2KΩ R28 2KΩ R29 2KΩ R30 2KΩ R31 2KΩ R32 2KΩ

C27 C28 C29 C30 C31 C32 C33 C34 C35 C36 C37 C38 C39 C40 C41 C42 C43 C44 C45 C46 C47 C48 620pF

C32 0.1uF C33 10uF C34 0.1uF C35 10uF C45 0.1uF C46 10uF

C49 22uF C50 1uF C51 1uF C52 1uF

+3.3V AGND

U2 ADS1298xPAG

IN1P IN1N IN2P IN2N IN3P IN3N IN4P IN4N IN5P IN5N IN6P IN6N IN7P IN7N IN8P IN8N

AVDD AVDD1 DVDD AVSS AVSS1 DGND

WCT RLDOUT RLDIN RLDINV RLDREF

TESTP_PACE_OUT1 TESTP_PACE_OUT2

GPIO1 GPIO2 GPIO3 GPIO4

DAISY_IN SCLK DIN CS DOUT

CLK CLKSEL START DRDY PWDN RESET RESV1

VCAP1 VCAP2 VCAP3 VCAP4 VREFP VREFN

VCAP1_ADC2 VCAP2_ADC2 VCAP3_ADC2 VCAP4_ADC2 VREFP_ADC2

SCLK MOSI CS_ADC2 MISO DRDY_ADC2 PWDN RESET

Unused, shorted

U1 CLK_EN=1 required

ADC_CLK +3.3V

DRDY: informational only; MCU still uses U1 DRDY

U3 ESP32-S3-DevKitC

CHIP_PU 5V 3V3 GND

GPIO43/U0TXD GPIO44/U0RXD

GPIO15/ADC2_CH4/32K_P GPIO16/ADC2_CH5/32K_N GPIO20/USB_D+ GPIO19/USB_D-

GPIO39/MTCK GPIO40/MTDO GPIO41/MTDI GPIO42/MTMS

GPIO0 GPIO1/ADC1_CH0 GPIO2/ADC1_CH1 GPIO3/ADC1_CH2 GPIO4/ADC1_CH3 GPIO5/ADC1_CH4 GPIO6/ADC1_CH5 GPIO7/ADC1_CH6 GPIO8/ADC1_CH7 GPIO9/ADC1_CH8 GPIO10/ADC1_CH9 GPIO11/ADC2_CH0 GPIO12/ADC2_CH1 GPIO13/ADC2_CH2 GPIO14/ADC2_CH3 GPIO17/ADC2_CH6 GPIO18/ADC2_CH7 GPIO21 GPIO35 GPIO36 GPIO37 GPIO38 GPIO45 GPIO46 GPIO47 GPIO48

RESET PWDN START DRDY_ADC1 CS_ADC1 MOSI SCLK MISO CS_ADC2

ESP32-S3-DevKitC

F21: probe access only; test-pad footprints/layout pending

TP1 ADC_CLK

TP2 DRDY_ADC1

TP3 DRDY_ADC2

TP4 3V3

TP5 BAT

TP6 AGND

TP7 VREFP_ADC1

TP8 VCAP1_ADC1

TP9 VCAP2_ADC1

TP10 VCAP3_ADC1

TP11 VCAP4_ADC1

TP12 VREFP_ADC2

TP13 VCAP1_ADC2

TP14 VCAP2_ADC2

TP15 VCAP3_ADC2

TP16 VCAP4_ADC2

Grounding Configuration

AGND AGND

ADC LDO Configuration

U4 LP2985-3.3

+3.7V C53 1uF EN VIN ON/OFF GND VOUT BP C55 10nF C54 2.2uF LP2985_OUT_DISABLED

F04: U4 output isolated with C54: no system load.
Both LDOs remain OFF. Qualify LP silicon/C54 before enable.

START R33 10k

RESET R34 10k

PWDN R35 10k

F09: 10k pulldowns bias START, RESET and PWDN low while MCU is high-Z.
CS defaults / complete power sequencing remain unresolved.

V1.2 PARTIAL CORRECTIONS — NOT READY FOR POWER-UP OR WEARER USE
F04: U4/C54 isolated from main rail. F05: R38 holds both LDOs OFF; operating power unqualified.
F06/F18: restore U1 CLK_EN after RESET; allow U2 clock/POR; configure before shared START.
F09: START/RESET/PWDN plus DIN/SCLK have pulldowns; CS/full sequencing unresolved.
Enable/settle 2.4V reference; verify gain/rate; keep PACE/RLD/WCT and RLD rerouting off.
F07/F08: protected body bias and wearer protection remain unimplemented.

Battery Configuration

+3.7V BT1 Battery AGND

MCU LDO Configuration

U5 TLV75533PDBV

+3.7V C56 1uF EN IN OUT GND C57 200uF +3.3V

MOSI R36 10k

SCLK R37 10k

EN R38 0

AGND

V1.2 F09: R36/R37 bias DIN and SCLK low; CS / full sequencing unresolved.
F05: R38 = OR FITTED: both LDOs held OFF. No functional enable source.
Keep R38 fitted until functional EN. F13/F15/F16/F17 qualification is complete.
USB / external +3.3V can still backfeed the rail; OFF strap is not isolation.

V1.2 F12/F14 — PROCUREMENT REQUIREMENTS (not qualified installed parts)
R1–R32: 2k unchanged; 0.1%, >=0.1W, derating/pulse review required.
All 32 input 620pF caps: C0G/NP0, 1%; electrode impedance/CMRR still unverified.
C25/C51 VCAP3: 1uF unchanged, >=10V; qualify capacitance at ~5.2V DC.
C55 BP: 10nF unchanged; leakage <100nA at operating bias/temperature.
No MPN, footprint, physical performance or wearer-safety qualification is implied.

Version 1.2
McGill AI Lab — Researcher: Karen Chen Lai
Sheet: /
File: open-semg-16-v1.3.kicad_sch
Title: sEMG Wirstband
Size: A2 | Date: | Rev:
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